

Contemporary C++: *Learning Modern C++ in a Modern Way*



الماس فناوری ابری پاسارگاد - آلفا

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Agenda 12/30

Session 5. Introduction to contiguously sequential data structures: C-Style arrays, C++ vectors, C++ standard arrays:

Part 1

- 📊 The C-Style array
- 📊 Array manipulation: writing simple programs
- 📊 Introduction to vectors
- 📊 References and further readings
- 📊 Q & A

120 min (incl. Q & A)



Contiguously sequential data structures

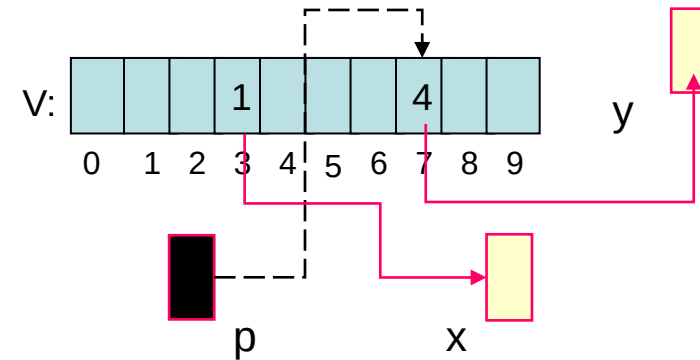
- Data structure
- Homogenous and Contiguous data structures
- Sequential data structures
- The most fundamental collection of data is a contiguously allocated sequence of elements of the same type, called an *array*.



Arrays

- For a type T, T[size] is the type “array of size elements of Type T.” The elements are indexed from 0 to size -1.
- [] is subscription operator.
- A C++ array is simply a sequence of memory locations.

```
int v[10]; // an array of 10 ints
v[3] = 1; // assign 1 to v[3]
int x = v[3]; // read from v[3]
int* p; // p is a pointer to an int
p = &v[7]; // assign the address of v[7] to p
*p = 4; // write to v[7] through p
int y = *p; // read from v[7] through p
```



- The number of array elements called **array bound**. Array bound is a constant and must be known at compile-time. If you need variable bound, use vector:

```
void f(int i)
{
    int v1[i]; // error: array size not a constant expression
    vector<int> v2(i); // ok
}
```

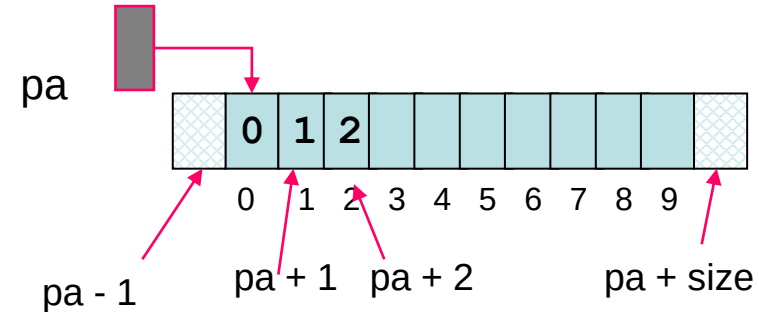
```
char vowel[] = { 'a', 'e', 'i', 'o', 'u' };
int a[5] = { 1, 2, 3, 4, 5 };
bool b[3] = { true, false };
```

- Array initializer:

Pointers, arrays and pointer arithmetic

- Pointers and arrays are closely related.

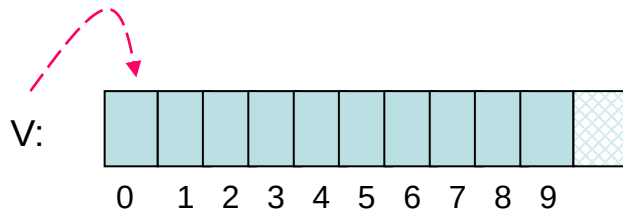
```
const int size = 10;  
*pa == 0; // OK  
*(pa + 1) == 1;  
*(pa + 2) == 2;  
*(pa + size) == one beyond the last element  
*(pa - 1) == undefined
```



- Taking a pointer to the element one beyond the end of an array is guaranteed to work.
- The name of an array is a pointer to the first element:

```
int v[10];      int *v;
```

```
int* p = &v[0];  
int* q = v
```



++ --
=*p *p=
+ - += -=
== !=
< <= >
>=
-> []

Pointer arithmetic operations

Arrays cont.

- Arrays don't know their sizes and bounds.

```
void fp(char v[], unsigned int size)
{
    for (int i = 0; i < size; i++)
        // use v[i]
}
```

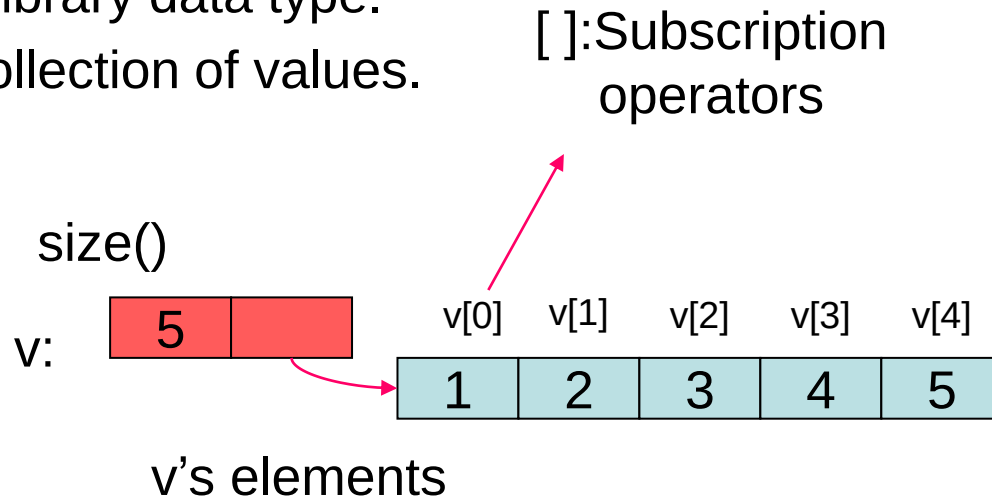
- Write a function that finds an integer value in the array. The function should return a pointer to the found element. If find failed, it should return pointer to one beyond the last element.

```
int* find(int v[], int vsize, int val) // find val in v
{
    for (int i = 0; i < vsize; i++)
        if (v[i] == val) return &v[i]; // if val is found return pointer to element
    return &v[vsize]; // if not found return pointer to one-beyond-the-end of v
}
```

Program

Vectors

- Vector is the most useful standard library data type.
- Vector is a **container** that holds a collection of values.
- All of the values in an individual vector are the same type, but different vectors can hold objects of different types.
- a `vector<T>` holds an sequence of values of type T.

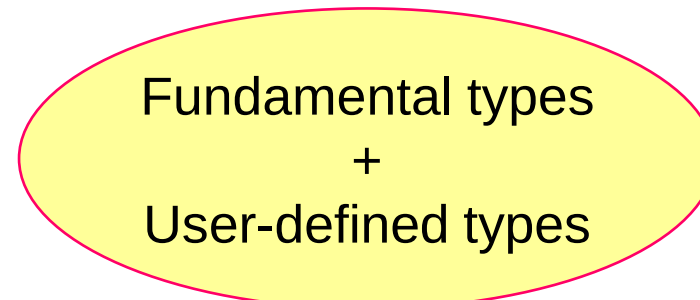


- examples:

`vector<int>`, `vector<double>`, `vector<string>`, `vector<vector<int> >`, ...

- The vector type is defined using language feature called **template classes**.

vector



- vector is defined in namespace std and in `<vector>` header file.

Vector cont.

- a Vector can grows as needed:

- `vector<int> v; // v is a vector of integer. start off empty`

V:

0	
---	--

- `v.push_back(1); // add an element with the value 1 at the end ("the back")`

V:

1	
---	--

 →

1

- `v.push_back(2); // add an element with the value 2 at the end ("the back")`

V:

2	
---	--

 →

1	2
---	---

- `v.push_back(3); // add an element with the value 3 at the end ("the back")`

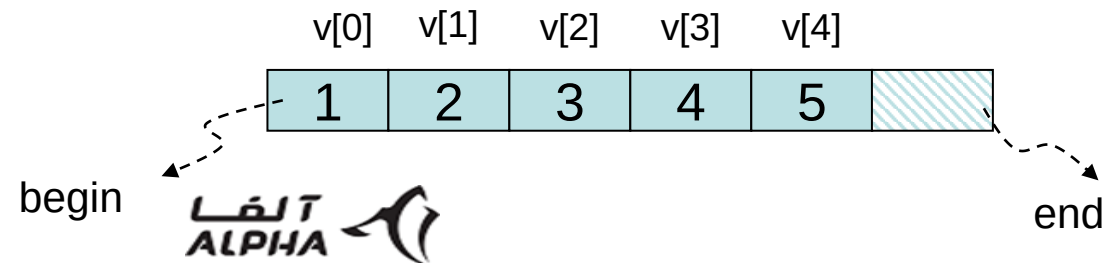
V:

3	
---	--

 →

1	2	3
---	---	---

- `push_back` pushes its argument onto the back of a vector. As a side effect, it increases the size of the vector by one.



R references and further readings

[KR78] Brian W. Kernighan & Dennis M. Ritchie. The C Programming Language, Prentice-Hall, Englewood Cliffs, NJ, 1978.

[Str00] Bjarne Stroustrup. The C++ Programming Language (Special Edition). Addison Wesley. Reading Mass. USA. February 2000.

[Str22] Bjarne Stroustrup. A Tour of C++, 3rd edition, Addison Wesley, September 2022.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

